ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Paper 2 style (design-related), built around one case study: an automatic sliding gate. Section A: objective questions (30) · Section B: the design process and system block diagram, torque, speed, power and gear calculations, relay control, motor reversing and battery back-up, safety, materials, maintenance and fault finding (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The motor of a sliding gate usually moves the gate with a …

A rack and pinion. B cam. C lever only. D crank.

1.1.2 The sensor that stops a closing gate when a car is in the way is the …

A thermometer. B infrared safety beam.
C microphone. D strain gauge.

1.1.3 The limit switches of a gate tell the controller when the gate is …

A fully open or fully closed. B wet.
C fully charged. D being opened by remote.

1.1.4 Torque is calculated as …

A force ÷ radius. B mass × speed.
C force × radius. D power × time.

1.1.5 A gate remote control sends its signal by …

A radio waves. B sound. C water. D magnetism only.

1.1.6 During a power failure, a gate motor keeps working from a …

A fuse. B resistor. C back-up battery. D capacitor only.

1.1.7 A relay contact rated at 10 A tells you the …

A coil resistance. B minimum voltage.
C switching speed. D maximum current it can safely switch.

1.1.8 The gear reduction in a gate motor gives …

A more speed and less torque. B more torque and less speed.
C the same speed and torque. D no movement.

1.1.9 Steel gate posts are best protected against rust by …

A heating. B magnetising. C galvanising or painting. D oiling only.

1.1.10 The first step of the design process is to …

A build the prototype. B identify the problem.
C test the product. D sell the product.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A safety beam must stop the gate when something blocks it.

1.2.2 Limit switches are actuators.

1.2.3 At the same motor speed, a larger pinion makes the gate move faster.

1.2.4 Power = work ÷ time.

1.2.5 A gate motor may run without any overload protection.

1.2.6 Galvanised steel resists corrosion.

1.2.7 A wiring diagram shows how components are connected.

1.2.8 The manual release lets the gate be moved by hand during a power failure.

1.2.9 A prototype is the final product that is sold to customers.

1.2.10 Gate motors never need maintenance.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Toothed bar fixed along the gate	A	torque
1.3.2	Small gear on the gearbox output shaft	B	limit switch
1.3.3	Tells the controller that the gate is fully open or closed	C	rack
1.3.4	Infrared beam across the driveway	D	prototype
1.3.5	Hand-held transmitter that opens the gate	E	remote control
1.3.6	Supplies power during load-shedding	F	manual release
1.3.7	Switches the motor off if it draws too much current	G	pinion
1.3.8	Turning effect of a force	H	back-up battery
1.3.9	First working model of a design	I	safety beam
1.3.10	Lets the gate be pushed by hand	J	overload protection

(10)

TOTAL SECTION A: [30]

SECTION B

CASE STUDY. The college needs an automatic sliding gate for the entrance of its robotics workshop. The 300 kg gate slides on wheels along a 4,5 m track. A 12 V DC motor with a gearbox drives a pinion on a rack fixed to the gate. The gate must open by remote control, stop at both ends, stop if a car is in the way, and work during load-shedding.

QUESTION 2: DESIGN PROCESS AND SYSTEM

2.1 Name the FIVE main steps of the design process in the correct order. (5)

2.2 Write FOUR specifications for the gate system. (4)

2.3 Draw a neat, labelled block diagram of the gate system showing the inputs, the controller and the outputs. (6)

2.4 Why should a warning light flash while the gate moves? (1)

2.5 Explain what the controller must do when the safety beam is broken while the gate is closing. (2)

[18]

QUESTION 3: MECHANICAL CALCULATIONS

3.1 A force of 400 N is needed to keep the gate moving. The pinion has a radius of 0,03 m. Calculate the torque needed at the pinion. (3)

3.2 The motor turns at 1 400 r/min and the gearbox ratio is 20 : 1. Calculate the speed of the pinion. (2)

3.3 Calculate the speed of the gate in m/s (pinion diameter 60 mm). (3)

3.4 Calculate how long the gate takes to open fully (4,5 m). (2)

3.5 Calculate the power needed to move the gate. (3)

3.6 Calculate the torque the motor must produce before the gearbox, ignoring losses. (2)

3.7 Explain why a gearbox is needed between the motor and the pinion. (3)

[18]

QUESTION 4: ELECTRICAL CONTROL

4.1 Draw a neat circuit diagram in which the controller switches a relay coil (with a flywheel diode), and the relay contacts switch the 12 V motor through a fuse. (6)

4.2 Explain how the direction of the DC motor can be reversed so that the gate can open and close. (3)

4.3 Explain how the limit switches are used to stop the motor at each end of the track. (3)

4.4 The back-up battery is rated 12 V, 7 Ah. Opening and closing the gate once draws 8 A for 20 s each way. Calculate how many complete open-and-close cycles the battery can supply. (4)

4.5 Choose a fuse rating for the 8 A motor circuit from 5 A, 10 A and 30 A and give a reason. (2)

[18]

QUESTION 5: SAFETY, MATERIALS AND MAINTENANCE

5.1 Give FOUR safety features that an automatic gate must have. (4)

5.2 Choose a suitable material for (a) the gate frame and (b) the rack, and give a reason for EACH. (4)

5.3 List FOUR maintenance tasks for the gate system. (4)

5.4 The gate stops halfway while opening. Give FOUR possible causes. (4)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER